



Assignment 3 — Prompting as an Engineering Practice

(Lecture: Prompt Lab + LangSmith Tracking)

Goal: Take one prompt through three measured, versioned iterations with evidence of quality lift — no “it feels better.”

Task

Build an **Employee Onboarding Assistant**: answers new-hire questions (benefits deadlines, IT setup, policies, who to contact). Warm but efficient, admits uncertainty, never invents policy.

Write a fake policy one-pager first (10–15 facts: enrollment deadline, laptop request process, PTO rule, etc.). This is your ground truth.

Define “good” before prompting: a 3–5 criterion rubric with scales, and an eval set of 8 questions — at least 2 easy, 2 ambiguous, 2 adversarial, and 2 whose answer is NOT in the policy doc (correct behavior = admit + escalate).

v1 → v2 → v3: write a first-draft system prompt, run it on all 8 questions, score against the rubric, then iterate twice — each revision must target specific logged failures, not general polish. Re-run the full eval set each time.

Instrument it: log all runs to LangSmith (traces tagged with version) — or, if you can’t get an account, keep an equivalent tracking spreadsheet: version, prompt text, model+params, per-question scores, timestamp.

Compare: produce a v1/v2/v3 score table per criterion. Identify at least one place a change *felt* better but the score didn’t move (or regressed).

Deliverables

Policy one-pager + rubric + 8-question eval set (dated/committed before v1 runs)

All 3 prompt versions + change notes (“v2 targets: hallucinated deadline in Q4, missing escalation in Q7”)

LangSmith project screenshots (or the tracking sheet)

Final comparison table + ½-page writeup: where did the lift actually come from, and what would your eval gate be before letting v4 ship?





Grading (100 pts)

Criterion	Pts
Rubric + eval set exist before v1 (evidenced) and cover easy/ambiguous/adversarial/unanswerable	2 5
Iterations target logged failures; same eval set reused each round	2 5
Tracking is complete: version, model, params, per-question scores tied to each run	2 0
Comparison is honest — includes the felt-vs-measured discrepancy	1 5
Ship-gate reasoning (what score bar, why)	1 5

Submission

Submit a single PDF or repo. Include model name/version and temperature for every run. Raw outputs must be unedited — cherry-picking a lucky run counts against the honesty criteria.

